EXERCISE · SLIDE 6 · QUESTION

PMF and PDF

Which values belong to the event?

Errors $-1, 0, 2$ cm have masses $1/2, 1/4, 1/4$. What is $P(x \geq 0)$?

A $1/4$: count only positive errors.

B $1/2$: include zero and positive errors.

C $2/3$: two of three possible values.

Event probability; PMF — [1] DL §3.3.1 · Course question

PMF and PDF

Which values belong to the event?

Solution 1 / 3

Identify the event

x: error (cm); $p(a) = P(x = a)$: probability mass at value a.

1. x: error in cm; $p(a) = P(x = a)$: probability mass at a.
2. Possible values: $-1, 0, 2$; masses: $1/2, 1/4, 1/4$.
3. Nonnegative means $x \geq 0$, so include 0 and 2.

Event probability; PMF — [1] DL §3.3.1 · Course question

EXERCISE · SLIDE 6 · SOLUTION 2/3

PMF and PDF

Which values belong to the event?

Solution 2 / 3

Add the included masses

x: error (cm); $p(a) = P(x = a)$: probability mass at value a.

---

1. $P(x \geq 0) = P(x = 0) + P(x = 2).$
2. $= 1/4 + 1/4 = 2/4 = 1/2.$
3. Check: $P(x < 0) + P(x \geq 0) = 1/2 + 1/2 = 1.$

Event probability; PMF — [1] DL §3.3.1 · Course question

EXERCISE · SLIDE 6 · SOLUTION 3/3

PMF and PDF

Which values belong to the event?

Solution 3 / 3

Choose B

x: error (cm); $p(a) = P(x = a)$: probability mass at value a.

---

1. A excludes zero and answers a different event.
2. B includes both nonnegative values and their masses.
3. C treats unequal masses as equal. Count probability, not values.

Event probability; PMF — [1] DL §3.3.1 · Course question

EXERCISE · SLIDE 6 · CHECK THE RESULT

PMF and PDF

Which values belong to the event?

Errors $-1, 0, 2$ cm have masses $1/2, 1/4, 1/4$. What is $P(x \geq 0)$?

A $1/4$: count only positive errors.

B $1/2$: include zero and positive errors. ✓

C $2/3$: two of three possible values.

> $P(x \geq 0) = p(0) + p(2) = 1/4 + 1/4 = 1/2$. Total mass is 1.

Event probability; PMF — [1] DL §3.3.1 · Course question

EXERCISE · SLIDE 9 · QUESTION

PMF and PDF

Density is height; probability is area

x is uniform on $[0, .5]$ cm. Which statement is valid?

A. Density 2 cm^{-1}; $P(.1 \leq x \leq .3) = .4$.

B. Density 2 means event probability 2.

C. Probability is .2, the interval width alone.

Density; event probability — [1] DL §3.3.2 · Course question

EXERCISE · SLIDE 9 · SOLUTION 1/3

PMF and PDF

Density is height; probability is area

Solution 1 / 3

Normalize the density

x: error (cm); h: uniform density; t: integration variable (cm).

1. x: error in cm; h: constant density on $[0, .5]$.
2. Total probability is area: $h(.5 - 0) = 1$.
3. $h = 1/.5 = 2\ \text{cm}^{-1}$.

Density; event probability — [1] DL §3.3.2 · Course question

PMF and PDF

Density is height; probability is area

Solution 2 / 3

Integrate over the requested interval

x: error (cm); h: uniform density; t: integration variable (cm).

1. $P(.1 \le x \le .3) = \int_{.1}^{.3} h \, dt.$
2. $= 2(.3 - .1) = 2(.2) = .4.$
3. Density times width has units $\text{cm}^{-1}\text{cm} = 1.$

Density; event probability — [1] DL §3.3.2 · Course question

EXERCISE · SLIDE 9 · SOLUTION 3/3

PMF and PDF

Density is height; probability is area

Solution 3 / 3

Choose A

x: error (cm); h: uniform density; t: integration variable (cm).

---

1. A gives both the normalized density and event probability.
2. B confuses density height with probability; probabilities cannot exceed 1.
3. C omits the height: width alone is not the area.

Density; event probability — [1] DL §3.3.2 · Course question

EXERCISE · SLIDE 9 · CHECK THE RESULT

PMF and PDF

Density is height; probability is area

x is uniform on $[0, .5]$ cm. Which statement is valid?

- **A** Density 2 cm^{-1}; $P(.1 \le x \le .3) = .4$. ✓
- **B** Density 2 means event probability 2.
- **C** Probability is .2, the interval width alone.

> $h(.5) = 1 \Rightarrow h = 2$; mass $= h(.3 - .1) = .4$.

Density; event probability — [1] DL §3.3.2 · Course question

EXERCISE · SLIDE 11 · QUESTION

Moments

What does centering remove?

$\mathrm{Var}(x) = \mathbb{E}[(x-\mu)^2]$. Given $\mu = 1/4$ cm and $\mathbb{E}[x^2] = 5/4$ cm^2, choose the variance.

A $5/4$: the second moment is always variance.

B 1: subtract the mean, not its square.

C $19/16$: subtract the squared mean.

Centering; variance — [1] DL §3.8 · Course question

Moments

What does centering remove?

Solution 1 / 3

Expand the centered square

x: error; $\mathbb{E}$: expectation; $\mu = \mathbb{E}[x]$: mean; Var: variance.

1. x: error; $\mu = \mathbb{E}[x] = 1/4$ cm.
2. $\mathbb{E}[x^2] = 5/4$ cm^2 is the second moment.
3. $(x - \mu)^2 = x^2 - 2\mu x + \mu^2$.

Centering; variance — [1] DL §3.8 · Course question

Moments

What does centering remove?

Solution 2 / 3

Take expectation, then substitute

x: error; $\mathbb{E}$: expectation; $\mu = \mathbb{E}[x]$: mean; Var: variance.

1. $\mathrm{Var}(x) = \mathbb{E}[x^2] - 2\mu\mathbb{E}[x] + \mu^2.$
2. $\mathbb{E}[x] = \mu \;\Rightarrow\; \mathrm{Var}(x) = \mathbb{E}[x^2] - \mu^2.$
3. $= 5/4 - (1/4)^2 = 20/16 - 1/16 = 19/16\ \mathrm{cm}^2.$

Centering; variance — [1] DL §3.8 · Course question

Moments

What does centering remove?

Solution 3 / 3

Choose C

x: error; $\mathbb{E}$: expectation; $\mu = \mathbb{E}[x]$: mean; Var: variance.

1. A leaves the squared mean inside the second moment.
2. B subtracts a length from a squared length; its units do not match.
3. C subtracts the squared mean and measures spread around the mean.

Centering; variance — [1] DL §3.8 · Course question

EXERCISE · SLIDE 11 · CHECK THE RESULT

Moments

What does centering remove?

$\mathrm{Var}(x) = \mathbb{E}[(x-\mu)^2]$. Given $\mu = 1/4$ cm and $\mathbb{E}[x^2] = 5/4$ cm^2, choose the variance.

A $5/4$: the second moment is always variance.

B 1: subtract the mean, not its square.

C $19/16$: subtract the squared mean. ✓

> $\mathrm{Var}(x) = \mathbb{E}[x^2] - \mu^2 = 5/4 - 1/16 = 19/16$ cm^2.

Centering; variance — [1] DL §3.8 · Course question

EXERCISE · SLIDE 16 · QUESTION

Moments

Independent sources, dependent components

U, V are independent fair signs. Let $x = (U, 2U + V)^T$. What is $\mathrm{Cov}(x_1, x_2)$?

A 2: both components share U.

B 0: independent sources imply independent outputs.

C 1: doubling a coefficient has no effect.

Independence; covariance — [1] DL §3.7 · [1] DL §3.8 · Course question

Moments

Independent sources, dependent components

Solution 1 / 4

Separate sources from components

U, V: independent fair signs; $x = (U, 2U + V)^T$; $\Sigma_{ij} = \mathrm{Cov}(x_i, x_j)$.

1. U, V: independent signs, each equally likely -1 or 1.
2. $\mathbb{E}[U] = \mathbb{E}[V] = (-1 + 1)/2 = 0$; hence $\mathbb{E}[x_1] = \mathbb{E}[x_2] = 0$.
3. $\mathbb{E}[U^2] = \mathbb{E}[V^2] = 1$; $\mathbb{E}[UV] = \mathbb{E}[U]\mathbb{E}[V] = 0$.

Independence; covariance — [1] DL §3.7 · [1] DL §3.8 · Course question

Moments

Independent sources, dependent components

Solution 2 / 4

Compute cross-covariance

U, V: independent fair signs; $x = (U, 2U + V)^T$; $\Sigma_{ij} = \mathrm{Cov}(x_i, x_j)$.

1. $\Sigma_{12} = \mathrm{Cov}(x_1, x_2) = \mathbb{E}[x_1 x_2]$.
2. $= \mathbb{E}[U(2U + V)] = 2\mathbb{E}[U^2] + \mathbb{E}[UV]$.
3. $= 2(1) + 0 = 2$.

Independence; covariance — [1] DL §3.7 · [1] DL §3.8 · Course question

Moments

Independent sources, dependent components

Solution 3 / 4

Check the complete covariance

U, V: independent fair signs; $x = (U, 2U + V)^T$; $\Sigma_{ij} = \mathrm{Cov}(x_i, x_j)$.

1. $\Sigma_{11} = \mathbb{E}[U^2] = 1.$
2. $\Sigma_{22} = \mathbb{E}[(2U + V)^2] = 4(1) + 4(0) + 1 = 5.$
3. $\Sigma = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}; \det \Sigma = 1(5) - 2(2) = 1.$

Independence; covariance — [1] DL §3.7 · [1] DL §3.8 · Course question

Moments

Independent sources, dependent components

Solution 4 / 4

Choose A

U, V: independent fair signs; $x = (U, 2U + V)^T$; $\Sigma_{ij} = \mathrm{Cov}(x_i, x_j)$.

1. A retains the shared source U.
2. B confuses independence of U and V with independence of their outputs.
3. C ignores the coefficient 2 in the shared contribution.

Independence; covariance — [1] DL §3.7 · [1] DL §3.8 · Course question

EXERCISE · SLIDE 16 · CHECK THE RESULT

Moments

Independent sources, dependent components

U, V are independent fair signs. Let $x = (U, 2U + V)^T$. What is $\mathrm{Cov}(x_1, x_2)$?

- **A** 2: both components share U. ✓
- **B** 0: independent sources imply independent outputs.
- **C** 1: doubling a coefficient has no effect.

> $\mathbb{E}[U(2U + V)] = 2$; $\Sigma = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$, $\det \Sigma = 1$.

Independence; covariance — [1] DL §3.7 · [1] DL §3.8 · Course question

EXERCISE · SLIDE 22 · QUESTION

Gaussian geometry

Which direction makes a point less surprising?

$\mu = 0, \Sigma = \mathrm{diag}(4, 1)$; $u = (2, 0)^T$, $v = (0, 2)^T$. Compare Gaussian densities.

A $p(u) = p(v)$: their Euclidean lengths agree.

B $p(u) > p(v)$: u is one standard deviation away.

C $p(u) < p(v)$: higher variance always lowers density.

Mahalanobis distance — [2] UBC Lec.18 p.5 · Course question

Gaussian geometry

Which direction makes a point less surprising?

Solution 1 / 3

Use the covariance-scaled distance

μ: mean; Σ: covariance; p: density; d_M: covariance-scaled distance.

1. $\mu = 0$: mean; $\Sigma = \mathrm{diag}(4, 1)$: covariance.
2. $u = (2, 0)^T$, $v = (0, 2)^T$ are points in the same Gaussian.
3. $d_M^2(x) = (x - \mu)^T \Sigma^{-1} (x - \mu) = x_1^2/4 + x_2^2$.

Mahalanobis distance — [2] UBC Lec.18 p.5 · Course question

Gaussian geometry

Which direction makes a point less surprising?

Solution 2 / 3

Compare exponents

μ: mean; Σ: covariance; p: density; d_M: covariance-scaled distance.

1. $d_M^2(u) = 2^2/4 + 0^2 = 1$; $d_M^2(v) = 0^2/4 + 2^2 = 4$.
2. $p(x) = c\exp(-d_M^2(x)/2)$; the normalization c is shared.
3. $p(u)/p(v) = e^{-1/2}/e^{-4/2} = e^{3/2} > 1$.

Mahalanobis distance — [2] UBC Lec.18 p.5 · Course question

Gaussian geometry

Which direction makes a point less surprising?

Solution 3 / 3

Choose B

μ: mean; Σ: covariance; p: density; d_M: covariance-scaled distance.

1. A uses ordinary distance and ignores different directional variances.
2. B: u is one standard deviation away; v is two.
3. C confuses changing covariance with comparing points in one fixed model.

Mahalanobis distance — [2] UBC Lec.18 p.5 · Course question

EXERCISE · SLIDE 22 · CHECK THE RESULT

Gaussian geometry

Which direction makes a point less surprising?

$\mu = 0, \Sigma = \mathrm{diag}(4, 1)$; $u = (2, 0)^T$, $v = (0, 2)^T$. Compare Gaussian densities.

A $p(u) = p(v)$: their Euclidean lengths agree.

B $p(u) > p(v)$: u is one standard deviation away. ✓

C $p(u) < p(v)$: higher variance always lowers density.

> $d_M^2(u) = 1, d_M^2(v) = 4; p(u)/p(v) = e^{3/2} > 1.$

Mahalanobis distance — [2] UBC Lec.18 p.5 · Course question

EXERCISE · SLIDE 28 · QUESTION

Bayes

Does a better base rate change the sensor?

$P(H) = .1$, $P(+ \mid H) = .9$, $P(+ \mid \neg H) = .1$. What is $P(H \mid +)$?

A .9: sensitivity is the positive predictive value.

B .1: an observation cannot change the prior.

C .5: true and false positives contribute equally.

Prior probability; posterior probability — [1] DL §3.11 · Course question

Bayes

Does a better base rate change the sensor?

Solution 1 / 3

Name the conditional populations

H: event; $\neg H$: no event; $+$: positive reading; P: probability.

1. H: event; $\neg H$: no event; $+$: positive test.
2. $P(H) = .1$, $P(\neg H) = 1 - .1 = .9$.
3. $P(+ \mid H) = .9$; $P(+ \mid \neg H) = .1$.

Prior probability; posterior probability — [1] DL §3.11 · Course question

Bayes

Does a better base rate change the sensor?

Solution 2 / 3

Find all positives, then normalize

H: event; $\neg H$: no event; $+$: positive reading; P: probability.

1. $P(H \cap +) = P(+ \mid H)P(H) = .9(.1) = .09.$
2. $P(\neg H \cap +) = P(+ \mid \neg H)P(\neg H) = .1(.9) = .09.$
3. $P(+) = .09 + .09 = .18$; $P(H \mid +) = .09/.18 = .5.$

Prior probability; posterior probability — [1] DL §3.11 · Course question

Bayes

Does a better base rate change the sensor?

Solution 3 / 3

Choose C

H: event; $\neg H$: no event; $+$: positive reading; P: probability.

1. A reverses the conditioning: sensitivity starts with event cases.
2. B uses the prior and ignores the positive observation.
3. C: true and false positives have equal probability mass.

Prior probability; posterior probability — [1] DL §3.11 · Course question

EXERCISE · SLIDE 28 · CHECK THE RESULT

Bayes

Does a better base rate change the sensor?

$P(H) = .1$, $P(+ \mid H) = .9$, $P(+ \mid \neg H) = .1$. What is $P(H \mid +)$?

A .9: sensitivity is the positive predictive value.

B .1: an observation cannot change the prior.

C .5: true and false positives contribute equally. ✓

$$P(H \mid +) = \frac{.9(.1)}{.9(.1) + .1(.9)} = .5.$$

Prior probability; posterior probability — [1] DL §3.11 · Course question

EXERCISE · SLIDE 34 · QUESTION

Gaussian conditioning

What does a noisier reading change?

Prior $\mathcal{N}(0, 4)$, reading $z = 3$. Noise variance rises from $R = 1$ to $R = 4$. What changes?

A. Mean moves toward 0; posterior variance increases.

B. Mean moves toward 3; posterior variance decreases.

C. Only the mean changes; variance ignores sensor noise.

Gaussian conditioning — [2] UBC Lec.18 p.7 · Course question

EXERCISE · SLIDE 34 · SOLUTION 1/3

Gaussian conditioning

What does a noisier reading change?

Solution 1 / 3

State the scalar measurement model

μ_0, P_0: prior mean/variance; μ_1, P_1: posterior mean/variance; K: measurement gain.

1. $x \sim \mathcal{N}(\mu_0, P_0)$ with $\mu_0 = 0$, $P_0 = 4$.
2. $z = x + \varepsilon = 3$; independent noise $\varepsilon \sim \mathcal{N}(0, R)$.
3. $K = P_0/(P_0 + R)$; $\mu_1 = \mu_0 + K(z - \mu_0)$; $P_1 = (1 - K)P_0$.

Gaussian conditioning — [2] UBC Lec.18 p.7 · Course question

Gaussian conditioning

What does a noisier reading change?

Solution 2 / 3

Compare the two noise variances

μ_0, P_0: prior mean/variance; μ_1, P_1: posterior mean/variance; K: measurement gain.

1. Old $R = 1$: $K = 4/5$, $\mu_1 = 0 + (4/5)3 = 2.4$, $P_1 = (1/5)4 = .8$.
2. New $R = 4$: $K = 4/8 = 1/2$, $\mu_1 = 0 + (1/2)3 = 1.5$.
3. New $P_1 = (1 - 1/2)4 = 2$; precision check: $P_1^{-1} = 1/4 + 1/4 = 1/2$.

Gaussian conditioning — [2] UBC Lec.18 p.7 · Course question

Gaussian conditioning

What does a noisier reading change?

Solution 3 / 3

Choose A

μ_0, P_0: prior mean/variance; μ_1, P_1: posterior mean/variance; K: measurement gain.

1. A: the mean is closer to 0 and posterior variance rises from .8 to 2.
2. B reverses both changes: a noisier reading carries less information.
3. C ignores R in the posterior variance. The sensor value z did not change.

Gaussian conditioning — [2] UBC Lec.18 p.7 · Course question

EXERCISE · SLIDE 34 · CHECK THE RESULT

Gaussian conditioning

What does a noisier reading change?

Prior $\mathcal{N}(0, 4)$, reading $z = 3$. Noise variance rises from $R = 1$ to $R = 4$. What changes?

- **A** Mean moves toward 0; posterior variance increases. ✓
- **B** Mean moves toward 3; posterior variance decreases.
- **C** Only the mean changes; variance ignores sensor noise.

> $K = 4/(4 + 4) = 1/2$; $\mu_1 = 1.5$, $P_1 = 2$ (before: 2.4, .8).

Gaussian conditioning — [2] UBC Lec.18 p.7 · Course question

EXERCISE · SLIDE 37 · QUESTION

Gaussian conditioning

Does the slice width change?

Gaussian $(a, b)^T$: $\mu = (1, 2)^T$, $\Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 9 \end{bmatrix}$. Observe $b = -1$ instead of 5.

A. Both conditional mean and variance decrease.

B. Mean decreases; conditional variance stays fixed.

C. Neither changes because the joint model is fixed.

Conditional Gaussian distribution — [2] UBC Lec.18 p.7 · Course question

Gaussian conditioning

Does the slice width change?

Solution 1 / 4

Read the Gaussian blocks

a, b: scalar components; μ: means; Σ: covariance blocks; $a \mid b$: a conditioned on b.

1. a: unobserved scalar; b: observed scalar.
2. $\mu_a = 1, \mu_b = 2; \Sigma_{aa} = 4, \Sigma_{ab} = \Sigma_{ba} = 2, \Sigma_{bb} = 9.$
3. $\mu_{a|b} = \mu_a + \Sigma_{ab}\Sigma_{bb}^{-1}(b - \mu_b).$

Conditional Gaussian distribution — [2] UBC Lec.18 p.7 · Course question

Gaussian conditioning

Does the slice width change?

Solution 2 / 4

Update the center

a, b: scalar components; μ: means; Σ: covariance blocks; $a \mid b$: a conditioned on b.

1. Old $b = 5$: $\mu_{a|b} = 1 + (2/9)(5 - 2) = 1 + 2/3 = 5/3$.
2. New $b = -1$: $\mu_{a|b} = 1 + (2/9)(-1 - 2) = 1 - 2/3 = 1/3$.
3. The innovation $b - \mu_b$ changed; the mean decreases.

Conditional Gaussian distribution — [2] UBC Lec.18 p.7 · Course question

Gaussian conditioning

Does the slice width change?

Solution 3 / 4

Compute the unchanged variance

a, b: scalar components; μ: means; Σ: covariance blocks; $a \mid b$: a conditioned on b.

1. $\Sigma_{a|b} = \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}.$
2. $= 4 - 2(1/9)2 = 36/9 - 4/9 = 32/9.$
3. This expression contains covariance blocks, not the observed value b.

Conditional Gaussian distribution — [2] UBC Lec.18 p.7 · Course question

Gaussian conditioning

Does the slice width change?

Solution 4 / 4

Choose B

a, b: scalar components; μ: means; Σ: covariance blocks; $a \mid b$: a conditioned on b.

1. A wrongly changes the width when only the observed value changes.
2. B changes the mean and keeps the conditional variance fixed.
3. C ignores the changed innovation. Constant width here uses joint Gaussianity.

Conditional Gaussian distribution — [2] UBC Lec.18 p.7 · Course question

EXERCISE · SLIDE 37 · CHECK THE RESULT

Gaussian conditioning

Does the slice width change?

Gaussian $(a, b)^T$: $\mu = (1, 2)^T$, $\Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 9 \end{bmatrix}$. Observe $b = -1$ instead of 5.

A. Both conditional mean and variance decrease.

B. Mean decreases; conditional variance stays fixed. ✓

C. Neither changes because the joint model is fixed.

> $\mu_{a|b} = 1 + \frac{2}{9}(-1 - 2) = \frac{1}{3}$; $\Sigma_{a|b} = 4 - \frac{4}{9} = \frac{32}{9}$.

Conditional Gaussian distribution — [2] UBC Lec.18 p.7 · Course question

EXERCISE · SLIDE 44 · QUESTION

Estimation

What makes the MAP estimate unique?

$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$, $b = (1, 2)^T$; prior center 0, $\lambda^2 = 1$. Why is MAP unique?

A. The prior makes the two data columns independent.

B. Gaussian noise makes every data fit unique.

C. The prior adds curvature along the data null space.

MAP estimate; positive definiteness — [1] DL §5.6.1 · Course question

Estimation

What makes the MAP estimate unique?

Solution 1 / 5

Identify the data null space

θ: parameter column; A: data matrix; b: data column; $r = b - A\theta$; I: 2×2 identity.

1. $\theta = (\theta_1, \theta_2)^T$: parameters; $b = (1, 2)^T$: readings.
2. $A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$; $r = b - A\theta$: residual.
3. $A\theta = (\theta_1 + \theta_2)(1, 2)^T$; zero residual requires $\theta_1 + \theta_2 = 1$.

MAP estimate; positive definiteness — [1] DL §5.6.1 · Course question

Estimation

What makes the MAP estimate unique?

Solution 2 / 5

Add the stated prior penalty

θ: parameter column; A: data matrix; b: data column; $r = b - A\theta$; I: 2×2 identity.

1. J: scaled negative-log posterior; $J = \frac{1}{2}\|r\|^2 + \frac{1}{2}\|\theta\|^2$, with $\lambda^2 = \sigma^2/\tau^2 = 1$.
2. $\nabla J = A^T(A\theta - b) + \theta = 0$.
3. $A^TA = \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix}$, $A^Tb = (5,5)^T$; hence $(A^TA + I)\theta = A^Tb$.

MAP estimate; positive definiteness — [1] DL §5.6.1 · Course question

Estimation

What makes the MAP estimate unique?

Solution 3 / 5

Solve the two equations

θ: parameter column; A: data matrix; b: data column; $r = b - A\theta$; I: 2×2 identity.

1. $6\theta_1 + 5\theta_2 = 5$; $5\theta_1 + 6\theta_2 = 5$.
2. Subtract: $\theta_1 - \theta_2 = 0$; substitute: $11\theta_1 = 5$.
3. $\hat{\theta}_{MAP} = (5/11, 5/11)^T$; $r = (1/11, 2/11)^T$.

MAP estimate; positive definiteness — [1] DL §5.6.1 · Course question

Estimation

What makes the MAP estimate unique?

Solution 4 / 5

Why the minimizer is unique

θ: parameter column; A: data matrix; b: data column; $r = b - A\theta$; I: 2×2 identity.

1. For any nonzero column v, $v^T(A^TA + I)v = \|Av\|^2 + \|v\|^2 > 0$.
2. For the data-null direction $v = (1, -1)^T$, $Av = 0$ but $\|v\|^2 = 2$.
3. Thus the total objective has positive curvature even where the data loss is flat.

MAP estimate; positive definiteness — [1] DL §5.6.1 · Course question

EXERCISE · SLIDE 44 · SOLUTION 5/5

Estimation

What makes the MAP estimate unique?

Solution 5 / 5

Choose C

θ: parameter column; A: data matrix; b: data column; $r = b - A\theta$; I: 2×2 identity.

1. A is false: A still has identical columns and rank 1.
2. B is false: Gaussian noise alone does not identify their separate coefficients.
3. C is correct: the proper Gaussian prior supplies curvature in the null direction.

MAP estimate; positive definiteness — [1] DL §5.6.1 · Course question

EXERCISE · SLIDE 44 · CHECK THE RESULT

Estimation

What makes the MAP estimate unique?

$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$, $b = (1, 2)^T$; prior center 0, $\lambda^2 = 1$. Why is MAP unique?

A. The prior makes the two data columns independent.

B. Gaussian noise makes every data fit unique.

C. The prior adds curvature along the data null space. ✓

> LS: $\theta_1 + \theta_2 = 1$. MAP: $(A^T A + I)\theta = A^T b$ gives $\theta_1 = \theta_2 = 5/11$.

MAP estimate; positive definiteness — [1] DL §5.6.1 · Course question

EXERCISE · SLIDE 45 · QUESTION

Integrated board

Calibration: choose the likelihood

$b = (1, 2, 3)^T$, $b_i = \theta + \varepsilon_i$, $\varepsilon_i \overset{iid}{\sim} \mathcal{N}(0, 1)$. Which data objective fits this model?

(A) $\frac{1}{2}\sum_i (b_i - \theta)^2$

(B) $\sum_i |b_i - \theta|$

(C) $\frac{1}{2}\sum_i (b_i - \theta)^2 + \frac{1}{2}\theta^2$

Gaussian likelihood; maximum likelihood — [1] DL §5.5.1 · Course example

Integrated board

Calibration: choose the likelihood

Solution 1 / 4

Define the calibration model

b_i: reading i; θ: offset; ε_i: noise; J_{data}: data objective; C: constant in θ.

1. $b = (1, 2, 3)^T$: observed readings; θ: unknown common offset.
2. $b_i = \theta + \varepsilon_i$; independent $\varepsilon_i \sim \mathcal{N}(0, 1)$.
3. $p(b_i \mid \theta) = (2\pi)^{-1/2} \exp[-(b_i - \theta)^2/2]$.

Gaussian likelihood; maximum likelihood — [1] DL §5.5.1 · Course example

EXERCISE · SLIDE 45 · SOLUTION 2/4

Integrated board

Calibration: choose the likelihood

Solution 2 / 4

From likelihood to objective

b_i: reading i; θ: offset; ε_i: noise; J_{data}: data objective; C: constant in θ.

1. Independence: $p(b \mid \theta) = \prod_{i=1}^{3} p(b_i \mid \theta)$.
2. $-\log p(b \mid \theta) = C + \frac{1}{2}\sum_{i=1}^{3}(b_i - \theta)^2$.
3. C is independent of θ; minimizing leaves $J_{data} = \frac{1}{2}\sum_i (b_i - \theta)^2$.

Gaussian likelihood; maximum likelihood — [1] DL §5.5.1 · Course example

EXERCISE · SLIDE 45 · SOLUTION 3/4

Integrated board

Calibration: choose the likelihood

Solution 3 / 4

Expand every square

b_i: reading i; θ: offset; ε_i: noise; J_{data}: data objective; C: constant in θ.

1. $2J_{data} = (1-\theta)^2 + (2-\theta)^2 + (3-\theta)^2.$
2. $= (1 - 2\theta + \theta^2) + (4 - 4\theta + \theta^2) + (9 - 6\theta + \theta^2).$
3. $= 3\theta^2 - 12\theta + 14.$

Gaussian likelihood; maximum likelihood — [1] DL §5.5.1 · Course example

Integrated board

Calibration: choose the likelihood

Solution 4 / 4

Minimize and choose A

b_i: reading i; θ: offset; ε_i: noise; J_{data}: data objective; C: constant in θ.

1. $J'_{data} = 3\theta - 6 = 0 \Rightarrow \hat{\theta}_{MLE} = 2$; $J''_{data} = 3 > 0$.
2. A follows from independent Gaussian noise.
3. B uses absolute residuals, a different noise model. C already adds a prior penalty.

Gaussian likelihood; maximum likelihood — [1] DL §5.5.1 · Course example

EXERCISE · SLIDE 45 · CHECK THE RESULT

Integrated board

Calibration: choose the likelihood

$b = (1, 2, 3)^T$, $b_i = \theta + \varepsilon_i$, $\varepsilon_i \overset{iid}{\sim} \mathcal{N}(0, 1)$. Which data objective fits this model?

- **A** $\frac{1}{2}\sum_i (b_i - \theta)^2$ ✓
- **B** $\sum_i |b_i - \theta|$
- **C** $\frac{1}{2}\sum_i (b_i - \theta)^2 + \frac{1}{2}\theta^2$

> $-\log p(b \mid \theta) = C + \frac{1}{2}\sum_i (b_i - \theta)^2$; $\hat{\theta}_{MLE} = \bar{b} = 2$.

Gaussian likelihood; maximum likelihood — [1] DL §5.5.1 · Course example

EXERCISE · SLIDE 46 · QUESTION

Integrated board

Calibration: combine data and prior

Same readings $(1, 2, 3)$; noise variance 1; prior $\theta \sim \mathcal{N}(0, 1)$. How does the posterior compare with the MLE?

A. Center stays at 2; prior changes only uncertainty.

B. Center moves toward 0; precision becomes 4.

C. Center moves toward 0; variances add to 4.

Gaussian posterior; MAP estimate — [2] UBC Lec.18 p.7 · [1] DL §5.6.1 · Course example

Integrated board

Calibration: combine data and prior

Solution 1 / 4

Keep the data and state the prior

θ: offset; J_{post}: posterior objective; μ_1, P_1: posterior mean/variance; precision $= 1/P_1$.

1. $b = (1, 2, 3)^T$; independent noise variance 1; θ: offset.
2. $J_{data} = \frac{1}{2}(3\theta^2 - 12\theta + 14)$; $\theta \sim \mathcal{N}(0, 1)$.
3. Negative log prior adds $\theta^2/2$, up to a constant.

Gaussian posterior; MAP estimate — [2] UBC Lec.18 p.7 · [1] DL §5.6.1 · Course example

Integrated board

Calibration: combine data and prior

Solution 2 / 4

Collect the posterior quadratic

θ: offset; J_{post}: posterior objective; μ_1, P_1: posterior mean/variance; precision $= 1/P_1$.

1. $J_{post} = \frac{1}{2}(3\theta^2 - 12\theta + 14) + \frac{1}{2}\theta^2.$
2. $= 2\theta^2 - 6\theta + 7 = 2(\theta^2 - 3\theta) + 7.$
3. $\theta^2 - 3\theta = (\theta - 3/2)^2 - 9/4.$

Gaussian posterior; MAP estimate — [2] UBC Lec.18 p.7 · [1] DL §5.6.1 · Course example

Integrated board

Calibration: combine data and prior

Solution 3 / 4

Complete the square and read variance

θ: offset; J_{post}: posterior objective; μ_1, P_1: posterior mean/variance; precision $= 1/P_1$.

1. $J_{post} = 2(\theta - 3/2)^2 - 9/2 + 7 = 2(\theta - 1.5)^2 + 2.5.$
2. Gaussian quadratic: $(\theta - \mu_1)^2/(2P_1)$; hence $\mu_1 = 1.5$, $1/(2P_1) = 2$.
3. $P_1 = 1/4$; precision $P_1^{-1} = 4$; $\theta \mid b \sim \mathcal{N}(1.5, 1/4)$.

Gaussian posterior; MAP estimate — [2] UBC Lec.18 p.7 · [1] DL §5.6.1 · Course example

Integrated board

Calibration: combine data and prior

Solution 4 / 4

Choose B and check the weights

θ: offset; J_{post}: posterior objective; μ_1, P_1: posterior mean/variance; precision $= 1/P_1$.

1. Data precision 3 plus prior precision 1 gives 4.
2. Posterior mean $= (3 \cdot 2 + 1 \cdot 0)/4 = 1.5$; MAP equals this Gaussian mean.
3. B is correct. A ignores the prior shift. C adds variances instead of precisions.

Gaussian posterior; MAP estimate — [2] UBC Lec.18 p.7 · [1] DL §5.6.1 · Course example

EXERCISE · SLIDE 46 · CHECK THE RESULT

Integrated board

Calibration: combine data and prior

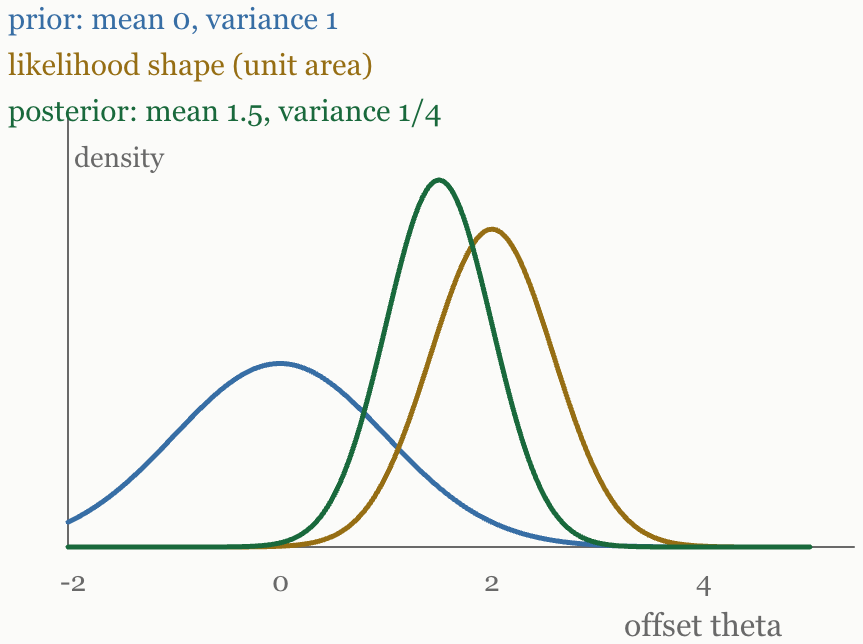
Same readings $(1, 2, 3)$; noise variance 1; prior $\theta \sim \mathcal{N}(0, 1)$.
How does the posterior compare with the MLE?

A. Center stays at 2; prior changes only uncertainty.

B. Center moves toward 0; precision becomes 4. ✓

C. Center moves toward 0; variances add to 4.

$J_{post} = 2(\theta - 1.5)^2 + 2.5; \theta \mid b \sim \mathcal{N}(1.5, 1/4).$



Gaussian posterior; MAP estimate — [2] UBC Lec.18 p.7 · [1] DL §5.6.1 · Course example

EXERCISE · SLIDE 47 · QUESTION

Intervals

Calibration: confidence coverage

Same sensor: $B_i \mid \theta \sim \mathcal{N}(\theta, 1)$ iid. $C(B) = \bar{B} \pm 1.96/\sqrt{3}$. What does 95% confidence mean?

A. This realized interval has 95% posterior mass.

B. 95% of future individual readings lie inside it.

C. Repeated intervals cover fixed θ 95% of the time.

Confidence interval; coverage probability — [3] NIST §7.1.4 · Course example

Intervals

Calibration: confidence coverage

Solution 1 / 4

Define the random interval

B: random data; $\bar{B}$: sample mean; θ: fixed offset; $C(B)$: confidence interval; Z: standardized mean.

1. θ: fixed true offset; B_i: random reading before observation.
2. $B_i \mid \theta \sim \mathcal{N}(\theta, 1)$ independently; $\bar{B} = (B_1 + B_2 + B_3)/3$.
3. $C(B)$: confidence interval computed anew for each random dataset.

Confidence interval; coverage probability — [3] NIST §7.1.4 · Course example

Intervals

Calibration: confidence coverage

Solution 2 / 4

Standardize the sample mean

B: random data; $\bar{B}$: sample mean; θ: fixed offset; $C(B)$: confidence interval; Z: standardized mean.

1. $\mathbb{E}_\theta[\bar{B}] = \theta$; $\mathrm{Var}_\theta(\bar{B}) = (1+1+1)/9 = 1/3$.
2. $Z = (\bar{B} - \theta)/(1/\sqrt{3}) \sim \mathcal{N}(0,1)$.
3. $P_\theta(-1.96 \le Z \le 1.96) \approx .95$.

Confidence interval; coverage probability — [3] NIST §7.1.4 · Course example

Intervals

Calibration: confidence coverage

Solution 3 / 4

Rearrange to isolate the fixed offset

B: random data; $\bar{B}$: sample mean; θ: fixed offset; $C(B)$: confidence interval; Z: standardized mean.

1. $-1.96/\sqrt{3} \leq \bar{B} - \theta \leq 1.96/\sqrt{3}.$
2. $\bar{B} - 1.96/\sqrt{3} \leq \theta \leq \bar{B} + 1.96/\sqrt{3}.$
3. Thus $C(B) = \bar{B} \pm 1.96/\sqrt{3}$ and $P_\theta(\theta \in C(B)) \approx .95.$

Confidence interval; coverage probability — [3] NIST §7.1.4 · Course example

Intervals

Calibration: confidence coverage

Solution 4 / 4

Choose C

B: random data; $\bar{B}$: sample mean; θ: fixed offset; $C(B)$: confidence interval; Z: standardized mean.

1. C: repeated random intervals cover the same fixed offset about 95% of the time.
2. A needs a posterior distribution; confidence alone provides none.
3. B concerns individual future readings, whose noise variance is 1, not 1/3.

Confidence interval; coverage probability — [3] NIST §7.1.4 · Course example

EXERCISE · SLIDE 47 · CHECK THE RESULT

Intervals

Calibration: confidence coverage

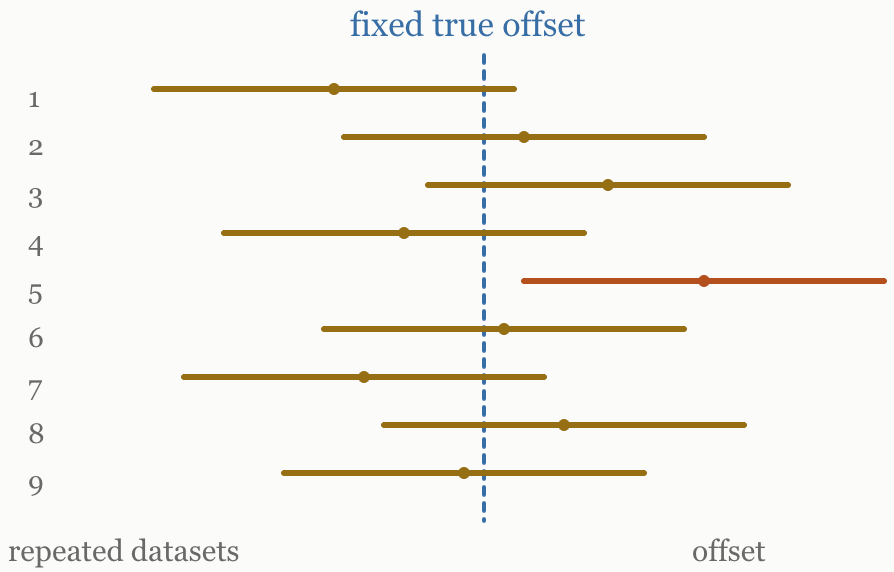
Same sensor: $B_i \mid \theta \sim \mathcal{N}(\theta, 1)$ iid. $C(B) = \bar{B} \pm 1.96/\sqrt{3}$. What does 95% confidence mean?

A. This realized interval has 95% posterior mass.

B. 95% of future individual readings lie inside it.

C. Repeated intervals cover fixed θ 95% of the time. ✓

$\sqrt{3}(\bar{B} - \theta) \sim \mathcal{N}(0, 1)$; $P_\theta(\theta \in C(B)) \approx .95$.



Illustration, not measured coverage

Confidence interval; coverage probability — [3] NIST §7.1.4 · Course example

EXERCISE · SLIDE 48 · QUESTION

Intervals

Calibration: what does credibility describe?

Observed $b = (1, 2, 3)^T$; $\theta \mid b \sim \mathcal{N}(1.5, .25)$. A 95% credible interval describes which probability?

A 95% posterior mass for θ, with b fixed.

B 95% coverage for every fixed true offset.

C 95% probability for a future noisy reading.

Posterior credible interval — [4] MIT 18.05 §2 · Course example

Intervals

Calibration: what does credibility describe?

Solution 1 / 3

Fix the data and use the posterior

b: fixed data; θ: offset; μ_1, P_1: posterior mean/variance; $I(b)$: credible interval; Z: standardized offset.

1. $b = (1, 2, 3)^T$: observed data, now fixed; θ: offset.
2. $\theta \mid b \sim \mathcal{N}(\mu_1, P_1)$ with $\mu_1 = 1.5$, $P_1 = .25$.
3. Posterior standard deviation $\sqrt{P_1} = \sqrt{.25} = .5$.

Posterior credible interval — [4] MIT 18.05 §2 · Course example

Intervals

Calibration: what does credibility describe?

Solution 2 / 3

Standardize, then undo the scaling

b: fixed data; θ: offset; μ_1, P_1: posterior mean/variance; $I(b)$: credible interval; Z: standardized offset.

1. $Z = (\theta - 1.5)/.5 \mid b \sim \mathcal{N}(0, 1).$
2. $P(-1.96 \leq Z \leq 1.96 \mid b) \approx .95.$
3. $1.5 - 1.96(.5) \leq \theta \leq 1.5 + 1.96(.5)$ gives $I(b) = [.52, 2.48].$

Posterior credible interval — [4] MIT 18.05 §2 · Course example

Intervals

Calibration: what does credibility describe?

Solution 3 / 3

Choose A

b: fixed data; θ: offset; μ_1, P_1: posterior mean/variance; $I(b)$: credible interval; Z: standardized offset.

1. A: $P(\theta \in I(b) \mid b) \approx .95$ under this posterior.
2. B is a repeated-sampling claim for every fixed offset; it does not follow here.
3. C includes new measurement noise; this interval describes the offset only.

Posterior credible interval — [4] MIT 18.05 §2 · Course example

EXERCISE · SLIDE 48 · CHECK THE RESULT

Intervals

Calibration: what does credibility describe?

Observed $b = (1, 2, 3)^T$; $\theta \mid b \sim \mathcal{N}(1.5, .25)$. A 95% credible interval describes which probability?

- **A** 95% posterior mass for θ, with b fixed. ✓
- **B** 95% coverage for every fixed true offset.
- **C** 95% probability for a future noisy reading.

> $I(b) = 1.5 \pm 1.96(.5)$; $P(\theta \in I(b) \mid b) \approx .95$.

Posterior credible interval — [4] MIT 18.05 §2 · Course example

EXERCISE · SLIDE 49 · QUESTION

Intervals

Calibration: why do the intervals differ?

Same data: $C(b) = 2 \pm 1.96/\sqrt{3}$; $I(b) = 1.5 \pm 1.96(.5)$. Why is I shifted and narrower?

A. It must have better repeated-sampling coverage.

B. The zero-centered prior adds information.

C. The physical sensor became less noisy.

Confidence interval; credible interval — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6.1 · Course comparison

Intervals

Calibration: why do the intervals differ?

Solution 1 / 3

Identify the two information sources

C: confidence interval; I: credible interval; $\bar{b}$: sample mean; precision: inverse variance.

1. $b = (1, 2, 3)^T$; $\bar{b} = 2$; noise variance 1; prior $\mathcal{N}(0, 1)$.
2. C: confidence interval; sampling standard error $1/\sqrt{3}$.
3. I: credible interval; posterior mean $(3 \cdot 2 + 1 \cdot 0)/4 = 1.5$, variance $1/4$.
4. Data precision $= 1/(1/3) = 3$; prior precision $= 1/1 = 1$.

Confidence interval; credible interval — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6.1 · Course comparison

Intervals

Calibration: why do the intervals differ?

Solution 2 / 3

Compute and compare endpoints

C: confidence interval; I: credible interval; $\bar{b}$: sample mean; precision: inverse variance.

1. $C = 2 \pm 1.96/\sqrt{3} \approx 2 \pm 1.1316 = [.8684, 3.1316]$.
2. $I = 1.5 \pm 1.96\sqrt{1/4} = 1.5 \pm .98 = [.52, 2.48]$.
3. $1.5 < 2$ and $.98 < 1.1316$: the prior shifts the center and reduces posterior width.

Confidence interval; credible interval — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6.1 · Course comparison

Intervals

Calibration: why do the intervals differ?

Solution 3 / 3

Choose B

C: confidence interval; I: credible interval; $\bar{b}$: sample mean; precision: inverse variance.

1. B: one unit of prior precision adds to three units from the data.
2. A: narrower does not prove better repeated-sampling coverage.
3. C: the physical noise variance remains 1; only the information used changed.

Confidence interval; credible interval — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6.1 · Course comparison

EXERCISE · SLIDE 49 · CHECK THE RESULT

Intervals

Calibration: why do the intervals differ?

Same data: $C(b) = 2 \pm 1.96/\sqrt{3}$; $I(b) = 1.5 \pm 1.96(.5)$. Why is I shifted and narrower?

A. It must have better repeated-sampling coverage.

B. The zero-centered prior adds information. ✓

C. The physical sensor became less noisy.

> $C \approx [.8684, 3.1316]$; $I \approx [.5200, 2.4800]$. Prior precision 1 adds to data precision 3.

Confidence interval; credible interval — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6.1 · Course comparison

EXERCISE · SLIDE 50 · QUESTION

Transfer

Calibration: independent transfer

New readings $b = (2, 3)^T$; iid noise $\mathcal{N}(0, 1)$; prior $\theta \sim \mathcal{N}(4, 1)$. Which conclusion is valid?

A. The prior changes the data-only MLE.

B. The prior center can be ignored in MAP.

C. MAP moves above the data mean; MLE stays fixed.

MLE; MAP; confidence and credible intervals — [1] DL §§5.5–5.6 · [3] NIST §7.1.4 · [4] MIT 18.05 §2 · Course question

Transfer

Calibration: independent transfer

Solution 1 / 5

Set up the new data-only problem

θ: offset; J: objective; μ_1, P_1: posterior mean/variance; C: confidence; I: credible interval.

---

1. $b = (2, 3)^T$; θ: offset; independent noise variance 1.
2. $J_{data} = \frac{1}{2}[(2-\theta)^2 + (3-\theta)^2]$.
3. $= \frac{1}{2}[(4 - 4\theta + \theta^2) + (9 - 6\theta + \theta^2)] = \frac{1}{2}(2\theta^2 - 10\theta + 13)$.

MLE; MAP; confidence and credible intervals — [1] DL §§5.5–5.6 · [3] NIST §7.1.4 · [4] MIT 18.05 §2 · Course question

EXERCISE · SLIDE 50 · SOLUTION 2/5

Transfer

Calibration: independent transfer

Solution 2 / 5

Minimize, then add the new prior

θ: offset; J: objective; μ_1, P_1: posterior mean/variance; C: confidence; I: credible interval.

---

1. $J'_{data} = 2\theta - 5 = 0 \Rightarrow \hat{\theta}_{MLE} = 2.5$; $J''_{data} = 2 > 0$.
2. Prior $\mathcal{N}(4, 1)$ adds $\frac{1}{2}(\theta - 4)^2 = \frac{1}{2}(\theta^2 - 8\theta + 16)$.
3. $J_{post} = \frac{1}{2}(3\theta^2 - 18\theta + 29)$.

MLE; MAP; confidence and credible intervals — [1] DL §§5.5–5.6 · [3] NIST §7.1.4 · [4] MIT 18.05 §2 · Course question

Transfer

Calibration: independent transfer

Solution 3 / 5

Complete the posterior square

θ: offset; J: objective; μ_1, P_1: posterior mean/variance; C: confidence; I: credible interval.

1. $3\theta^2 - 18\theta + 29 = 3(\theta^2 - 6\theta) + 29.$
2. $= 3[(\theta - 3)^2 - 9] + 29 = 3(\theta - 3)^2 + 2.$
3. $J_{post} = \frac{3}{2}(\theta - 3)^2 + 1;\ \mu_1 = 3,\ 1/(2P_1) = 3/2 \Rightarrow P_1 = 1/3.$

MLE; MAP; confidence and credible intervals — [1] DL §§5.5–5.6 · [3] NIST §7.1.4 · [4] MIT 18.05 §2 · Course question

Transfer

Calibration: independent transfer

Solution 4 / 5

Compute both interval constructions

θ: offset; J: objective; μ_1, P_1: posterior mean/variance; C: confidence; I: credible interval.

1. C: confidence; mean 2.5, sampling standard error $1/\sqrt{2}$.
2. $C = 2.5 \pm 1.96/\sqrt{2} \approx [1.1141, 3.8859]$.
3. I: credible; posterior mean 3, SD $1/\sqrt{3}$; $I = 3 \pm 1.96/\sqrt{3} \approx [1.8684, 4.1316]$.

MLE; MAP; confidence and credible intervals — [1] DL §§5.5–5.6 · [3] NIST §7.1.4 · [4] MIT 18.05 §2 · Course question

EXERCISE · SLIDE 50 · SOLUTION 5/5

Transfer

Calibration: independent transfer

Solution 5 / 5

Choose C

θ: offset; J: objective; μ_1, P_1: posterior mean/variance; C: confidence; I: credible interval.

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1. C: $\hat{\theta}_{MAP} = 3 > 2.5$, while the data-only MLE remains 2.5.
2. A incorrectly puts prior information into the data-only MLE.
3. B drops the prior center 4 and its linear term. Confidence and credibility retain different meanings.

MLE; MAP; confidence and credible intervals — [1] DL §§5.5–5.6 · [3] NIST §7.1.4 · [4] MIT 18.05 §2 · Course question

EXERCISE · SLIDE 50 · CHECK THE RESULT

Transfer

Calibration: independent transfer

New readings $b = (2, 3)^T$; iid noise $\mathcal{N}(0, 1)$; prior $\theta \sim \mathcal{N}(4, 1)$. Which conclusion is valid?

A The prior changes the data-only MLE.

B The prior center can be ignored in MAP.

C MAP moves above the data mean; MLE stays fixed. ✓

> $\hat{\theta}_{MLE} = 2.5, \theta \mid b \sim \mathcal{N}(3, 1/3)$. $C = 2.5 \pm 1.96/\sqrt{2}$; $I = 3 \pm 1.96/\sqrt{3}$.

MLE; MAP; confidence and credible intervals — [1] DL §§5.5–5.6 · [3] NIST §7.1.4 · [4] MIT 18.05 §2 · Course question

EXERCISE · SLIDE 51 · QUESTION

Intervals

What does the sensor promise?

A specification says “±2 cm, 95%.” Which conclusion is justified without another definition?

A. It is a confidence interval for the offset.

B. It is a credible interval for the offset.

C. None: ask what quantity and probability model it describes.

Confidence; credibility; predictive uncertainty — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6, Eq.5.68 · Course question

Intervals

What does the sensor promise?

Solution 1 / 3

Identify what the specification omits

$C(B)$: confidence interval; $I(b)$: credible interval; ε_{new}: future-reading error.

1. The statement "± 2 cm, 95%" supplies a width and a percentage.
2. θ: offset; B: repeated random dataset; b: observed dataset.
3. No target quantity or probability construction is specified.

Confidence; credibility; predictive uncertainty — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6, Eq.5.68 · Course question

Intervals

What does the sensor promise?

Solution 2 / 3

Compare three possible meanings

$C(B)$: confidence interval; $I(b)$: credible interval; ε_{new}: future-reading error.

1. Confidence: $P_\theta(\theta \in C(B)) \approx .95$ for a stated repeated-data procedure.
2. Credibility: $P(\theta \in I(b) \mid b) \approx .95$ under a stated posterior.
3. Future-error bound: $P(|\varepsilon_{new}| \leq 2 \text{ cm}) \approx .95$ under a stated error model.

Confidence; credibility; predictive uncertainty — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6, Eq.5.68 · Course question

Intervals

What does the sensor promise?

Solution 3 / 3

Choose C

$C(B)$: confidence interval; $I(b)$: credible interval; ε_{new}: future-reading error.

1. A assumes an unstated repeated-sampling procedure.
2. B assumes an unstated prior and posterior.
3. C: ask which quantity is bounded and which model or procedure supplies 95%.

Confidence; credibility; predictive uncertainty — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6, Eq.5.68 · Course question

EXERCISE · SLIDE 51 · CHECK THE RESULT

Intervals

What does the sensor promise?

A specification says “±2 cm, 95%.” Which conclusion is justified without another definition?

A. It is a confidence interval for the offset.

B. It is a credible interval for the offset.

C. None: ask what quantity and probability model it describes. ✓

> Offset uncertainty, future-reading error and repeated coverage are different claims.

Confidence; credibility; predictive uncertainty — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6, Eq.5.68 · Course question

EXERCISE · SLIDE 52 · QUESTION

Synthesis

Which modeling bridge is valid?

With iid Gaussian noise and an isotropic Gaussian prior, which modeling chain is valid?

A. Independence gives squares; Gaussian noise gives sums.

B. Gaussian noise gives squares; independence gives sums.

C. A Gaussian prior removes the data residual term.

Gaussian likelihood; independence; MAP regularization — [1] DL §§5.5–5.6 · [1] DL §3.7 · Course question

Synthesis

Which modeling bridge is valid?

Solution 1 / 4

State the model and symbols

i: reading index; I: identity matrix; C: constant in θ; J: negative-log posterior, up to constants.

1. $b = A\theta + \varepsilon$: data column b, design matrix A, parameter column θ.
2. ε_i are independent $\mathcal{N}(0, \sigma^2)$ noises; $r = b - A\theta$: residual.
3. Prior $\theta \sim \mathcal{N}(\mu_0, \tau^2 I)$; μ_0: prior mean; $\sigma^2, \tau^2 > 0$: variances.

Gaussian likelihood; independence; MAP regularization — [1] DL §§5.5–5.6 · [1] DL §3.7 · Course question

Synthesis

Which modeling bridge is valid?

Solution 2 / 4

Separate the square from the sum

i: reading index; I: identity matrix; C: constant in θ; J: negative-log posterior, up to constants.

1. One Gaussian factor: $p(b_i \mid \theta) = (2\pi\sigma^2)^{-1/2} e^{-r_i^2/(2\sigma^2)}$.
2. Independence gives $p(b \mid \theta) = \prod_i p(b_i \mid \theta)$.
3. $-\log p(b \mid \theta) = C + \sum_i r_i^2/(2\sigma^2) = C + \|b - A\theta\|^2/(2\sigma^2)$.

Gaussian likelihood; independence; MAP regularization — [1] DL §§5.5–5.6 · [1] DL §3.7 · Course question

Synthesis

Which modeling bridge is valid?

Solution 3 / 4

Add the prior through Bayes rule

i: reading index; I: identity matrix; C: constant in θ; J: negative-log posterior, up to constants.

1. $p(\theta \mid b) \propto p(b \mid \theta)p(\theta)$; logarithms turn this product into a sum.
2. $J = \|b - A\theta\|^2/(2\sigma^2) + \|\theta - \mu_0\|^2/(2\tau^2)$.
3. Multiply all of J by σ^2: $\frac{1}{2}\|r\|^2 + \frac{\lambda^2}{2}\|\theta - \mu_0\|^2$, where $\lambda^2 = \sigma^2/\tau^2$.

Gaussian likelihood; independence; MAP regularization — [1] DL §§5.5–5.6 · [1] DL §3.7 · Course question

Synthesis

Which modeling bridge is valid?

Solution 4 / 4

Choose B

i: reading index; I: identity matrix; C: constant in θ; J: negative-log posterior, up to constants.

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1. B: the Gaussian exponent supplies squares; independence and logarithms supply the sum.
2. A reverses the two reasons.
3. C discards the likelihood. A prior adds a penalty while retaining the residual term.

Gaussian likelihood; independence; MAP regularization — [1] DL §§5.5–5.6 · [1] DL §3.7 · Course question

EXERCISE · SLIDE 52 · CHECK THE RESULT

Synthesis

Which modeling bridge is valid?

With iid Gaussian noise and an isotropic Gaussian prior, which modeling chain is valid?

A. Independence gives squares; Gaussian noise gives sums.

B. Gaussian noise gives squares; independence gives sums. ✓

C. A Gaussian prior removes the data residual term.

$$-\log p(\theta \mid b) = C + \|b - A\theta\|^2/(2\sigma^2) + \|\theta - \mu_0\|^2/(2\tau^2).$$

Gaussian likelihood; independence; MAP regularization — [1] DL §§5.5–5.6 · [1] DL §3.7 · Course question